Assessing the Association of Genetics, Cognitive and General Function, and Imaging Biomarkers with Risk of Progression from Mild Cognitive Impairment to Dementia

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0.1 Abstract:

In this report we examine how genetics, cognitive and general function, and imaging biomarkers are associated with risk of progression from mild cognitive impairment (MCI) to dementia, in Alzheimer's Disease (AD). MCI is often the first clinical sign of AD, but there is large variation in whether participants progress from MCI to dementia, and how quickly they do. Improving ability to assess participants' relative risk of progression has implications in the allocation of medical resources and in helping individuals and their families prepare for the future. We use data from the Alzheimer's Disease Neuroimaging Initiative (ADNI), a long-term study measuring biomarkers of AD. Survival analysis methods are used to estimate the effects of genetics (APOE4 allele status), cognitive and functional scores (ADAS13, CDRSB, MMSE, ADNI-EF, ADNI-Mem scores), and imaging biomarkers (amyloid plaque PET and ventricle, hippocampus, and entorhinal cortex MRI images) on risk of progression from MCI to dementia. Using a Cox proportional hazards model, we find a concordance of 0.88 using a model fit on amyloid presence, hippocampus volume, and ADAS13, CDRSB, and ADNI-EF scores, and years of education. In smaller models, the most powerful predictor of risk of progression, by AIC and concordance, was ADAS13, followed by ADNI-Mem, and CDRSB score. Further studies should assess the value of combinations of these predictors in terms of information on relative risk of participants, weighed against costs of large-scale applications. Our conclusions are qualified by a portion of missing genetics and imaging data in the data set.

0.2 Introduction

Alzheimer's disease (AD) is a neuro-degenerative disease characterized by a buildup of amyloid plaque and other proteins in the brain, the atrophy of certain regions of the brain, and a decline in mental functioning over time. In its end stages, AD often leads to debilitating dementia. AD is the leading cause of dementia globally and presents a major challenge for global health.

Early treatment of AD may be the most effective way of managing the disease. However the early stages of AD are poorly understood. Memory impairment is often the first clinical sign of the disease, with a diagnosis of mild cognitive impairment (MCI). The path from MCI to dementia varies largely between individuals. Progression may take less than a year or more than a decade. The prevalence of MCI is about 7% at the ages of 60-64 years, and increases to about 25% by the ages of 80-84 years. (New Facts & Figures Report on Mild Cognitive Impairment | Mayo Clinic Connect, n.d.). Cognitive and functional scores, genetic data, and imaging of the brain are important metrics in the characterization of AD. The effect of these metrics on progression from MCI to AD is not yet fully understood.

Functional and cognitive scores attempt to quantify a participant's memory, thinking, and reasoning skills. These tests often involve questions or assessments of ability to complete small tasks. Many have been developed and used in the assessment of dementia. The AD Assessment Scale (ADAS13), a test of cognitive and mental ability, was developed for the assessment of later stages of dementia and has since been used widely in pre-dementia assessments. High ADAS13 scores indicate worse function. The Clinical Dementia Rating Score – Sum of Boxes (CDRSB), is another score derived from an interview assessing cognitive and functional abilities. CDRSB is widely used in assessing early-stage AD. (Nyunt et. al 2013). High CDRSB scores indicate worse function. The Mini Mental State Exam (MMSE), is an 11-question assessment that tests cognitive function. It is relatively easy and cheap to administer and has been in use since the 1970s. (Geriatric Nursing Resources, n.d.). High MMSE scores indicate better function. ADNI-EF and ADNI-Mem are two composite scores for assessing cognition that have been developed by ADNI to synthesize the most informative questions used in other cognitive scores. ADNI-EF focuses on characterizing executive function and ADNI-Mem on memory. They were developed specifically to assess cognitive decline in pre-dementia participants, and to standardize other scores. (Gibbons et al., 2012) (Crane et al., 2012). High ADNI-EF and ADNI-Mem scores indicate better function. These two scores are generally standard normally distributed. All these measurements of cognitive function are in use in the assessment of MCI and dementia, and we aim to use the ADNI dataset to better quantify their relative impact on risk of progression from MCI to dementia.

The pathology of AD involves changes to the structure of the brain. A variety of brain imaging techniques have emerged as useful indicators of AD pathology. The development of amyloid plaques in the brain is one of the most definite clinical signs of AD (Chapleau et al., 2022). Amyloid PET scans provide a quantification of the amount of amyloid in the brain – those who are deemed amyloid positive have substantial amyloid burden in the brain. Magnetic resonance imaging (MRI) is used to visualize regions of the brain linked to the

AD. The hippocampus and the entorhinal cortex are two regions of the brain associated with memory, and their atrophy is a step in the pathology of AD. Ventricles in the brain will swell as the other regions shrink and thus are also well used indicators of AD. We wish to quantify the effect that changes in these imaging biomarkers has on the relative hazard of dementia.

In recent years the role of genetics in AD has been increasingly investigated. The APOE4 allele is the biggest genetic risk factor for late onset Alzheimer's disease, so E4 carriers (having at least one allele) are thought to be at greater risk of progressing to dementia.

The purpose of this study is to use ADNI data to estimate how each of these commonly used metrics affect the risk of progressing from MCI to dementia.

0.3 Methods

0.3.1 Data Overview

The Alzheimer's Disease Neuroimaging Initiative (ADNI) is a large multi-site study that began in 2004, with the primary goal of identifying markers of AD (Weiner et al. 2024). This report will use data from individuals recruited as part of ADNI-GO, ADNI-2, or ADNI-3. The individuals in this dataset all entered the study with MCI. The participants were drawn from some 50 sites across the US and Canada. The participants in this dataset were between the ages of 55 and 90 when they entered the study. Upon entering the study, baseline information on genetics, cognitive function, and brain images were measured. The participants were then monitored for progression to dementia or until the end of the study.

The dataset used in this study contains 16 variables. Time, in years, to progression to dementia is the response of interest. An indicator tells whether the time was censored. Gender (male/female), years of education (integer), and age (integer) were collected at baseline. Imaging variables measured at baseline include amyloid presence (Positive or Negative), and the volumes of the ventricle, hippocampus and entorhinal regions in the brain (integers). Genetic information consists of the participant's number of APOE4 alleles (0,1, or 2). Five different functional and cognitive metrics (CDRSB, ADAS13, MMSE, ADNI_Mem, and ADNI_EF) are also included.

Of the 604 observations in the data set, 159 observations have at least one missing predictor value. Missing measurements are present in the following variables: Age (3), APOE4 (40), ADAS13 (2), Ventricles (44), Hippocampus (72), Entorhinal (77), and amyloid status (6). Between all reported observations and all complete cases, means and standard deviations of numeric variables remain similar and categorical compositions are relatively unchanged. We assume that the data is missing completely at random. In this case, observations with missing variables can be dropped without introducing bias into the data set.

Table 1: Data Overview

	All Cases	Complete Case
Obsevations	604.0	445.0
Mean Age	71.9	71.5
SD Age	7.5	7.3
% Age (0,60]	5.3	6.1
% Age (60,65]	15.0	16.0
% Age (65,70]	19.8	20.7
% Age (70,75]	23.5	22.7
% Age (75,80]	22.3	22.2
% Age (80,100)	14.1	12.4
Mean Education	16.2	16.2
SD Education	2.6	2.6
% Male	56.6	55.5
Mean years	3.8	3.9
SD years	2.9	2.9
% dementia	27.5	28.5
% amyloid	53.5	54.4
% No Allel	52.3	53.0
% One Allel	37.4	37.1
% Two Alleles	10.3	9.9
Mean CDRSB	1.5	1.5
SD CDRSB	0.9	0.9
Mean ADAS13	15.1	14.6
SD ADAS13	6.7	6.4
Mean MMSE	28.0	28.0
SD MMSE	1.8	1.8
Mean Ventricle Vol	38206.2	37202.1
SD Ventricle Vol	21431.1	20688.4
Mean Hippocampus Vol	7090.1	7101.0
SD Hippocampus Vol	1098.8	1090.0
Mean Entorhinal Vol	3736.5	3731.0
SD Entorhinal Vol	757.3	734.9
Mean ADNI-MEM	0.4	0.4
SD ADNI-MEM	0.7	0.7
Mean ADNI-EF	0.4	0.5
SD ADNI-EF	0.8	0.8

We examine collinearity in categorical variables through Pearson chi-square testing. In numerical predictors we examine VIF to assess collinearity. There is a significant correlation between the number of APOE4 alleles and the presence of amyloid plaques in a participant (Pearson's Chi-squared test p-value $<$ 2.2e-16). An increasing number of APOE4 alleles correlates with increased prevalence of amyloid positive individuals. Models containing

both amyloid and APOE4 status may have inflated standard error estimates and degraded interpretability, so a model containing both should be avoided. All numeric variables have a VIF between 1 and 2, so we do not detect strong collinearity in those variables.

Table 2: Categorical Pearson Tests of Correlation.

	Amyloid_negative	Amyloid_positive
Zero	196	95
One	54	157
Two	7	50

0.3.2 Methods of Analysis

We assess the association of the variables of interest with progression from MCI to dementia through survivor analysis techniques. A Cox regression is constructed to estimate the effects of the biomarkers. To construct a Cox model, we assume that censoring is uninformative. Cox models also assume the hazards of predictors are proportionally constant over time. We investigate nonproportionality of risk through Kaplan-Meier (KM) and Nelson-Anderson (NA) survival curve estimations, as well as through Schoenfeld residuals of built Cox models.

Non-Parametric analysis

KM and NA survival curves estimate the survival function for progression to dementia based on the number of alleles of APOE4 and the presence of amyloid plaques at baseline. We construct the KM survival curves, compare them, and test for differences. We construct the cumulative hazard curves using NA cumulative hazard estimates and compare them. We use NA estimates of cumulative hazard to create hazard ratios between allele statuses and between amyloid statuses. We use these estimates to establish a difference in hazard function between these groups and to test for the proportionality of hazards.

Cox modeling

We fit an initial Cox regression to estimate the influence of demographic variables. From this initial model, we add biomarkers to estimate the proportional risk of changes in predictor values and evaluate the ability to predict the relative risk of participants using concordance.

Initial fit

We initially regress survival times on our three demographic variables, gender, education, and age. Schoenfeld residuals of this model are used to assess the significance of the non-proportionality of the hazard functions. Martingale residuals are interpreted to assess the necessity of changes in the functional form of numeric variables. We consider stratification, modification of functional form, and introduction of interaction effects as mediations of potential non-proportionality. Non-significant predictors are dropped from the model, and the reduced and full initial fit are tested using ANOVA or a Wald test (depending on the number of predictors dropped) before proceeding to add additional predictors.

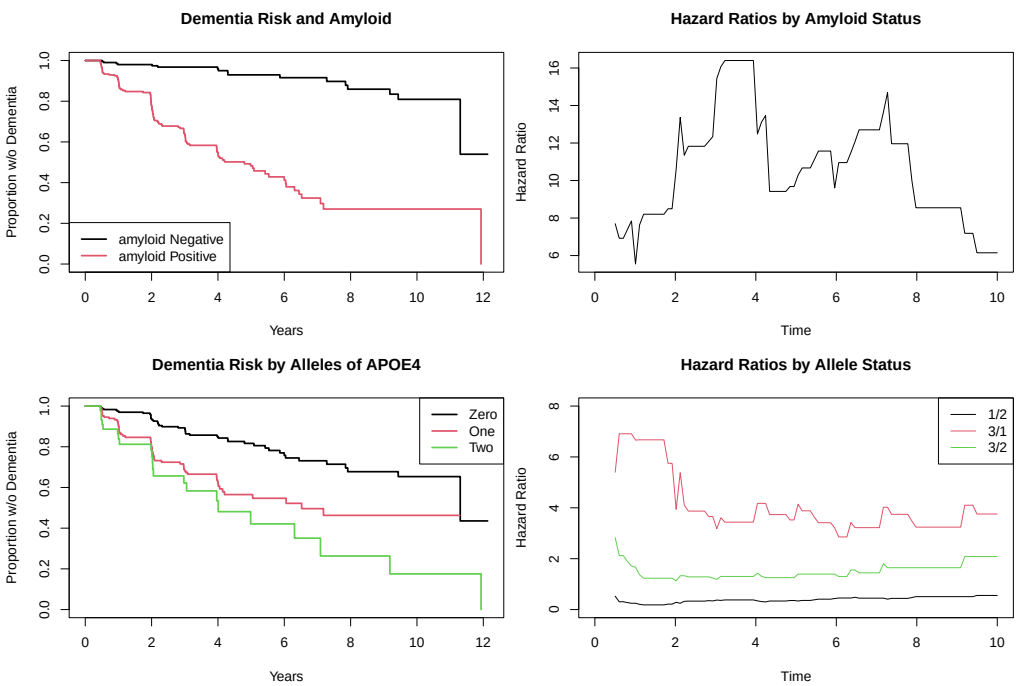
Residuals are plotted to identify any unusual data points and examine them. Model wide martingale residuals and deviance residuals are plotted against the linear predictor. Dfbeta residuals of individual predictors are plotted against observation order. Points identified in multiple plots are examined and interpreted using the coefficients of the model. Obvious errors may be excluded from the final model.

A full model is created by introducing the remaining biomarkers. The examination of functional form and proportionality through residuals, as described for the initial model, is repeated for the full model. Residuals are plotted, as in the initial fit, to discover influential or outlying points.

We also construct smaller models, by adding single predictors to the initial model. From these models we assess single predictors effect on risk of progression, single predictor's ability to predict relative risk of individuals, and compare to the coefficients to those from the full model.

A reduced model is formed from the full model by dropping predictors one by one based on significance. Additional predictors are then dropped one by one if they lower AIC by 2 or more. The achieved model is then compared to the full model using LRT.

0.4 Results



The KM survival estimate shows a lower survival curve for the amyloid positive group. A Mantel-Haenszel test between the amyloid statuses gives a p-value < 2x10^-16. We can reject the null hypothesis that the two amyloid statuses have the same survival function. Plots of survival and cumulative hazards over time do not appear nonparallel, so we do not see any evidence of non-proportionality. The ratio of amyloid positive to negative hazard fluctuates but remains a similar magnitude and is constant in direction. There remains no

strong evidence of non-proportional hazards in amyloid status

The KM survival estimate shows the lowest survival curve for the carriers of two APOE4 alleles, followed by one, and then zero. A Mantel-Haenszel test between the alleles' survival curves gives a p value = 1x(10^-8). We can reject the null hypothesis that the APOE4 allele statuses all have the same survival function. Plots of survival and cumulative hazards over time do not appear nonparallel, so we do not see any evidence of non-proportionality. The ratio of allele statuses are stable over time. There remains no strong evidence of non-proportional hazards in APOE4 status

0.4.1 Model Building outcomes

A Cox model of survival times is constructed using age, education, and gender. Schoenfeld residuals of age indicate significant non-proportionality. Martingale residuals for age do not show a clear transformation. A model with interaction terms between age and the other predictors also shows significant non proportionality through the age Schoenfeld residual, so interaction terms are not used to alleviate proportionality. For stratification based on age, we create a new age variable, where age is categorical, with age groups: (0,60], (60,65], (65,70], (70,75], (75,80], (80,100].

A second model is fit on education and gender and stratified by age category. Stratification in a Cox model assumes the coefficients for predictors are the same across all strata. To test this assumption, we run an ANOVA between the stratified model, and one with terms for interactions between strata and predictors. This ANOVA has a p-value of 0.69. We fail to reject the null hypothesis that all strata interaction terms are 0. This indicates the strata interactions do not significantly improve the model. There is no evidence that stratification by age is an inappropriate method of addressing non proportionality in this model.

In the stratified model, Schoenfeld residuals do not indicate further non-proportionality. Martingales residuals for education do not indicate the necessity for a transformation. Education is significant. Gender has little effect and is not significant, so it is dropped. Unusual points identified with martingale, deviance, and dfbeta residuals did not reveal any obvious errors. A full breakdown of these points is found in the appendix.

After adding all the biomarkers to the initial model, residuals do not suggest issues with proportionality or functional form. Unusual points identified by martingale, deviance, and dfbeta residuals do not reveal obvious errors. A full breakdown of these points is found in the appendix.

A final reduced model is produced by dropping biomarkers based on significance by individual Wald tests. Predictors that can be dropped without lowering AIC or concordance are then also dropped. The final model includes the following predictors: education, amyloid status, hippocampus volume, ADAS13 score, CDRSB score, ADNI-EF score, with age category as strata. A Chi-squared likelihood ratio test between this model and the full model has a test statistic with p-value is 0.12. We do not reject the null hypothesis that all dropped predictors have a coefficient of zero.

0.4.2 Model Characteristics

Table 3: Individual Risk Estimates.

Predictor	B	L95 e^(B)	U95 e^(B)	indv. e^(B)	std. e^(B)*	Concord	AIC
Ventricles	0.000	1.00	1.00	1.00	1.35	0.596	941.9
Hippocampus	-0.001	1.00	1.00	1.00	0.48	0.722	892.1
Entorhinal	-0.001	1.00	1.00	1.00	0.56	0.657	916.3
Amyloid +	2.328	5.88	17.90	10.26	10.26	0.723	849.2
1 APOE4	1.012	1.84	4.11	2.75	2.75	0.663	914.2
2 APOE4	1.554	2.75	8.12	4.73	4.73	0.663	914.2
CDRSB	0.732	1.79	2.42	2.08	2.00	0.776	877.2
ADAS13	0.181	1.16	1.23	1.20	3.33	0.795	813.9
MMSE	-0.236	0.72	0.86	0.79	0.66	0.647	927.7
ADNI_EF	-0.969	0.29	0.50	0.38	0.38	0.717	900.2
ADNI_MEM	-1.822	0.11	0.23	0.16	0.16	0.781	826.3

The table above gives information from models built by adding single predictors to the initial model (education and age strata.) Columns include predictor's effects on risk of progression from MCI to dementia, as well as concordance and AIC of the models containing them in this dataset. Column B shows the predictors' effect on the linear predictor. e^B gives the proportional change in risk of dementia for a one unit change in the predictor. For APOE4 alleles, the estimates are comparisons of risk between that number of alleles and no APOE4 alleles. For amyloid, we have the proportional risk of amyloid positive individuals to negative. To improve comparability, the column "stdz e^B" shows a standardized version of "e^B". For this estimate, the variables ADAS13, CDRSB, MMSE, ventricle volume, hippocampus volume, and entorhinal volume were standardized using sample mean and standard deviation. This helps produce a comparison between these predictors whose ranges and variations are different. Caution should be used when interpreting these standardized coefficients, as this assumes a normal distribution in these predictors, which is likely not the case for some predictors (see distributions in appendix 1).

The individual model estimates an exponentiated standardized coefficient for hippocampus volume of 0.48. This means that every one unit increase in standardized hippocampus volume (1090.003 units) corresponds to 48% the hazard of progressing to dementia, at any given time, not accounting for other covariates. For ventrical volume, the exponentiated standardized coefficient is 1.35. This means that a one standardized-unit increase in ventricle volume (20688.45) corresponds to 135% the hazard of progressing to dementia, at any given time, not accounting for other covariates. Similarly, a standardized-unit increase (734.9364 units) in entorhinal value corresponds to 56% the hazard of the original value. The estimated exponentiated coefficent for amyloid positive is 10.26. Therefore changing from negative to positive amyloid status increases hazard to 1026% its original value. Similarly moving from zero APOE4 alleles to one increases hazard to 275% its original value, and a change from zero alleles to two increases hazard to 473% the original hazard. A participant with a one unit

increase in CDRSB or ADAS13, would have a hazard 208% or 120% its original estimate, respectively. A participant with a one unit increase in MMSE, ADNI-EF, ADNI-Mem would have a hazard 66%, 38%, or 16% its original estimate, respectively. All of these estimated changes in hazard only take into account one variable.

Table 4: Final Model Risk Estimates

Predictor	B	L95 e^(B)	U95 e^(B)	e^(B)	std. e^(B)*
Hippocampus	-0.0004	1.00	1.00	1.0	0.6121
Amyloid +	1.9029	3.76	11.97	6.7	6.7055
CDRSB	0.1045	1.07	1.15	1.1	2.0042
ADAS13	0.5551	1.46	2.08	1.7	1.6915
ADNI_EF	-0.4420	0.49	0.85	0.6	0.6428

The final model contains hippocampus volume, amyloid status and ADAS13, CDRSB and ADNI-EF scores. The concordance of the model is 89%. This level of concordance suggests the model fits the data fairly well. The full model and the reduced final model are not distinguishable in terms of likelihood, concordance, or AIC. The estimates in the e^(B) columns of this table are the estimated proportional change in risk from a value change in the selected predictor, all other predictors in that model held constant.

The model estimates a coefficient for hippocampus volume of -0.0004. This means that every one unit increase in hippocampus volume corresponds to a 0.0004 lower log hazard of progressing to dementia, at any given time, all other predictors held constant. So a one unit increase in hippocampus volume results in 99.96% of the hazard. The model estimates a coefficient for amyloid positive of 1.90. This means that an amyloid positive participant has a 6.7 times higher hazard of progressing to dementia than a amyloid negative participant, at any given time, all other predictors held constant. The model estimates a coefficient for CDRSB score of 0.105. This means that every one unit increase in CDRSB score corresponds to a 0.105 increase in the log hazard of progressing to dementia, at any given time, all other predictors held constant. So a one unit increase in CDRSB score results in 111.0% of the hazard. The model estimates a coefficient for ADAS13 score of 0.555. This means that every one unit increase in ADAS13 score corresponds to a 0.555 increase in the log hazard of progressing to dementia, at any given time, all other predictors held constant. So a one unit increase in ADAS13 score results in 174% of the hazard. The model estimates a coefficient for ADNI-EF score of -0.442. This means that every one unit increase in ADNI-EF score corresponds to a 0.442 decrease in the log hazard of progressing to dementia, at any given time, all other predictors held constant. So a one unit increase in ADNI-EF score results in 64.3% of the hazard.

0.5 Discussion:

Individually each of the measurements are helpful indicators of risk. Estimates of risk from the full model are not included due to issues with overfitting and collinearity. The estimates

from models including only one biomarker give an estimate of the relative risk of a change in just that value, without accounting for other biomarkers. As such, these estimates do not function in estimating the effect of multiple changes in predictors. They do allow for an assessment for the impact of a single variable on the relative risk of progression from MCI to dementia, and in comparing relative implication of singular measurements. For all functional and cognitive scores, scores associated with better mental function are associated with a decreased relative risk of dementia. Similarly, MRI volumes estimated coefficients are in the correct direction. The concordance of the smaller models is meaningful in how well these single predictors can help explain variation in risk across participants in this dataset. For instance, based on these concordances, ADNI-Mem explains variation in participants' hazards better than ventricle volumes in this dataset.

From the final reduced model, we can see a subset of predictors investigated in this study that are together helpful in assessing the relative risk of individuals. In all cases these estimates are closer to 1 than the corresponding estimate from a single predictor model. For instance, moving from amyloid negative vs amyloid positive, a single predictor model estimates the proportional change in risk is above 10, while in the final model that estimate is 6.7, other predictors held constant. One explanation for this pattern is collinearity between the predictors. The exclusion of ventricle volumes, entorhinal volumes, APOE4 alleles, MMSE scores, and ADNI-EF scores from the final model means that, in this dataset, having already accounted for the other variables, these predictors do not significantly improve the ability to estimate participants' relative risks.

The value of cognitive and functional scores is underlined by this study. Individually functional and cognitive scores performed well compared to the rest of the biomarkers. ADNI-MEM, CDRSB, and ADAS13 have the three best individual concordances. ADAS13 and ADNI-MEM also were the two best individual fits to the data by AIC. Three of the five scores, ADAS13, CDRSB, and ADNI-EF, remained in the final model. This model has a good fit to the data with a concordance of 0.89. Knowing more than one score is helpful in explaining variation in risk in this data set. The MMSE model had the worst fit on the data of all the cognitive and functional scores. It also was not included in the final model.

Imaging techniques were also valuable in assessing participants' relative risks. The remaining two predictors in our final model were amyloid plaque presence and hippocampus volume. Amyloid positive participants had an estimate 6.7 times the risk of developing dementia (95% CI [3.8,12]), compared to otherwise identical amyloid negative participants. Decreasing hippocampus size significantly increased the risk of dementia. Decreased entorhinal volumes also caused increased risk; however, it fits the risk seen in this data set less well than hippocampus volumes. Ventricles showed a further decline from entorhinal volume, based on AIC and concordance.

The number of APOE4 alleles was not significant in the final model. This may be due to its strong correlation with amyloid plaque status (Pearson chi square test p value $< 2 \times 10$ ^-16). In the final model, if we substitute amyloid status for APOE4 alleles, the model fits the data well, but less well than the final model, with concordance of 0.87 and AIC of 734 (compared to 0.89 and 699 in the final model). The number of APOE4 alleles individually are also poor predictors of participants' relative risk, compared to other predictors used in

this study. In this study genetics were comparatively poor predictors in assessing participant risk of progression from MCI to dementia.

The influence of age on the risk of progression is important. However, this study does not directly estimate how age affects risk. Stratifying on age prevents direct estimation. Interactions between age and other variables were not detected in this study but remain possible. Given the availability of participant age and its large influence on risk, further studies should address how age interacts with the effects of cognitive and functional metrics, imaging biomarkers, and genetics on risk of dementia.

Further studies should assess the most informative combinations of predictors. Our reduced final model fits the data well but involves multiple tests requiring differing domain expertise to administer. Gathering this information for a large number of people is difficult and expensive. Single biomarker models have fit this data set less well (concordance of .67-.79 vs 0.89). Further study could identify the best fitting combination of 2 or 3 predictors in a Cox regression.

Further studies should also focus on investigating more predictors. We see in our final model that adding additional imaging and score-based predictors helps improve the fit of the model. There are many other imaging and score-based indicators used in the monitoring of AD that may help improve risk assessment. The study of additional genetic markers will be of use as well.

0.5.1 Limitations

Cox models assume that censoring is non-informative. This assumption is not verifiable from this data and may impact the validity of the results of this study. Long-term studies, such as this one are vulnerable to drop out induced bias, which is possible here.

The implications of this study are limited to populations similar to that sampled in ADNI. ADNI sampling techniques deliberately exclude participants with comorbidities . The estimates of risk assessed in this study are more applicable to people solely affected by AD. The interaction of other cognitive conditions is not assessed here.

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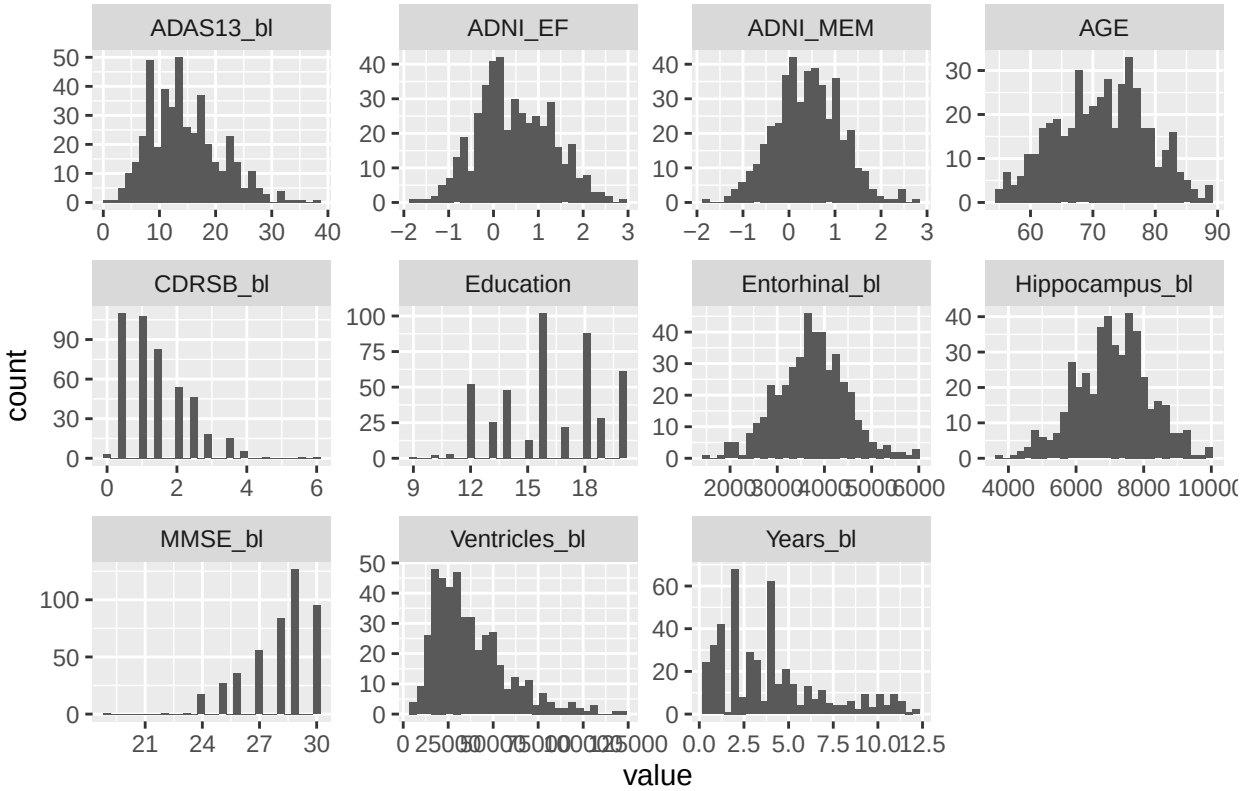
2 *Appendix 1*

2.1 EDA

Continuous Predictor Histograms



Categorical Predictor Bargraphs

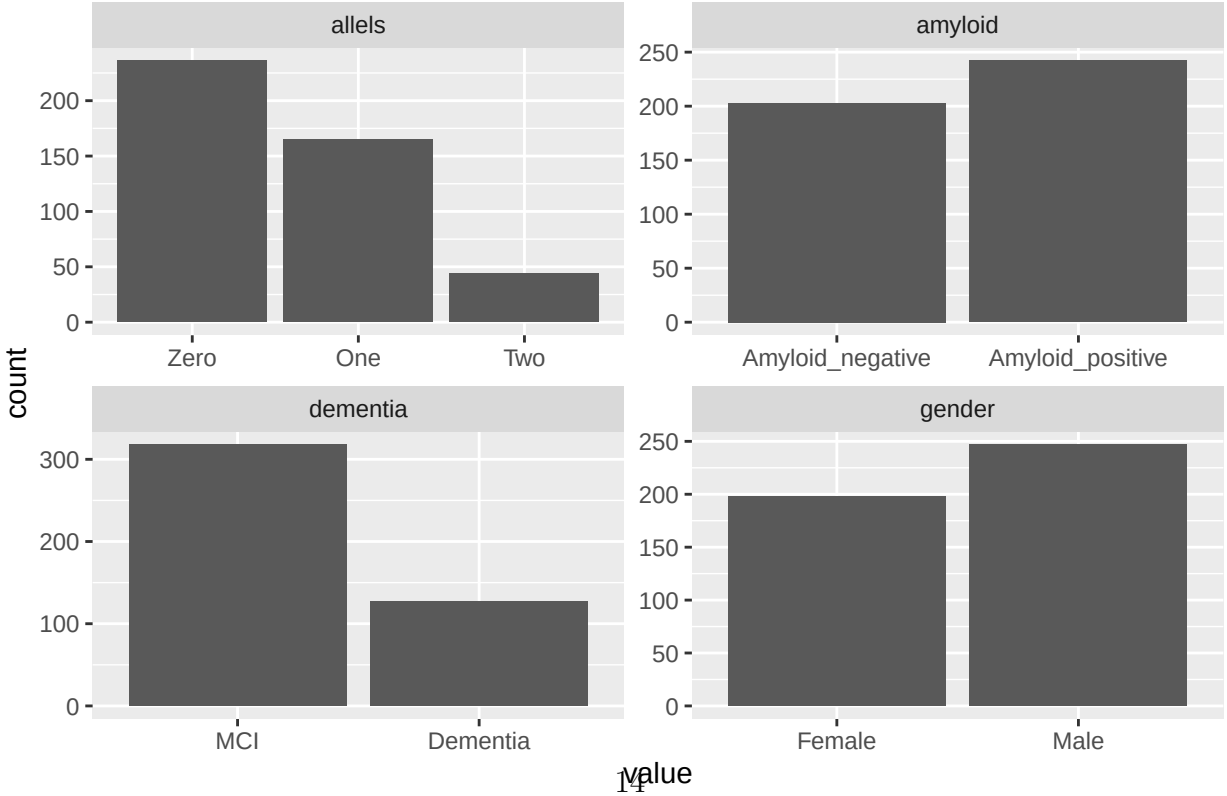


Table 6: Categorical Pearson Tests of Correlation.

	Zero	One	Two
Female	127	93	24
Male	168	118	34

	Amyloid_negative	Amyloid_positive
Female	118	143
Male	160	177

Table 5: Categorical Pearson Tests of Correlation.

	Amyloid_negative	Amyloid_positive
Zero	196	95
One	54	157
Two	7	50

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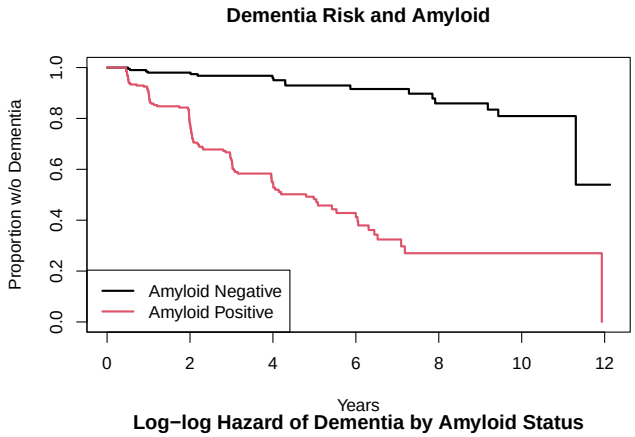
##
##  Pearson's Chi-squared test
##
## data:  table(data_full$allels, data_full$amyloid)
## X-squared = 114.9, df = 2, p-value < 2.2e-16
```

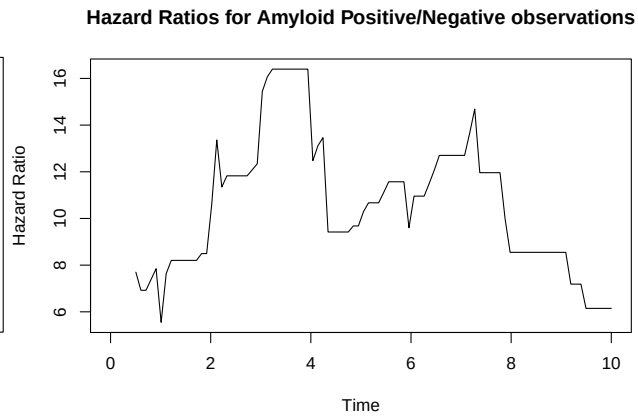
	MCI	Dementia
Female	188	74
Male	250	92

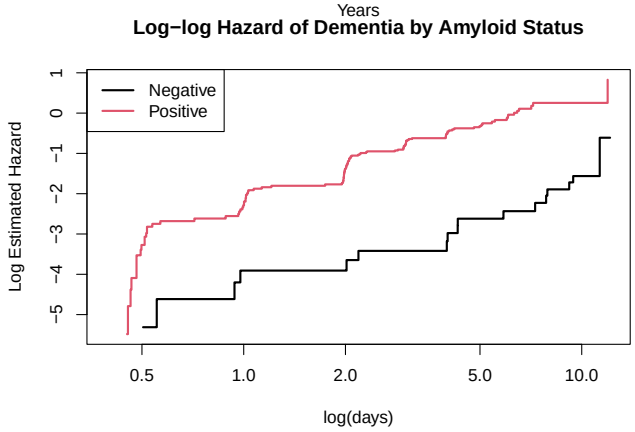
	MCI	Dementia
Amyloid_negative	254	24
Amyloid_positive	180	140

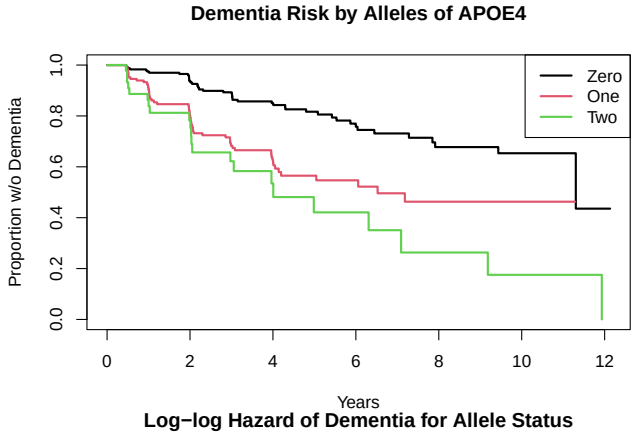
	MCI	Dementia
Zero	240	55
One	133	78
Two	32	26

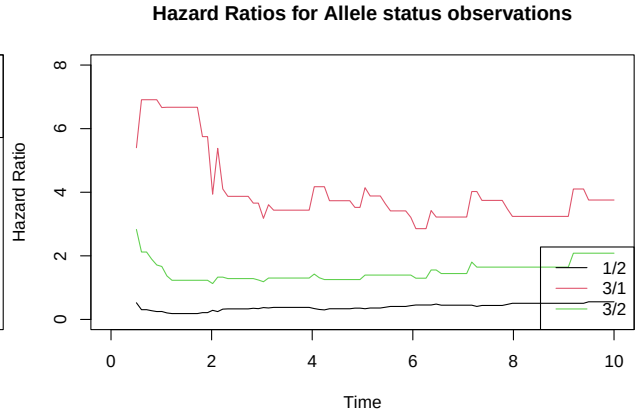
2.2 Survival Curves

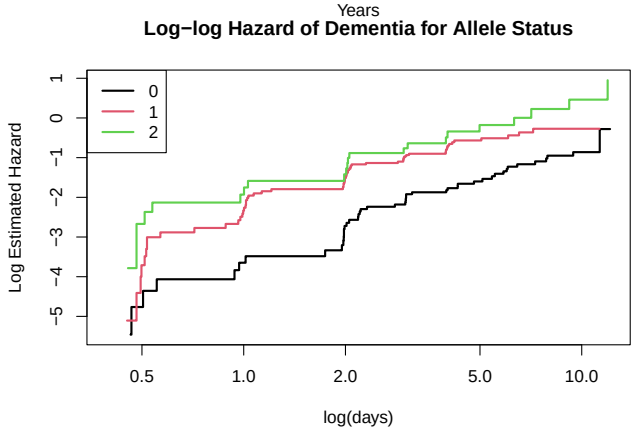




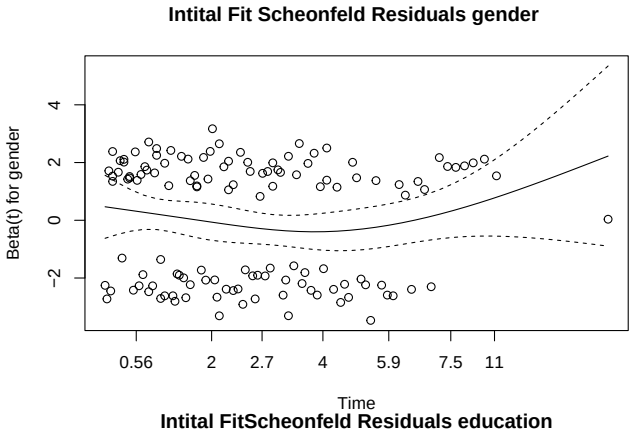


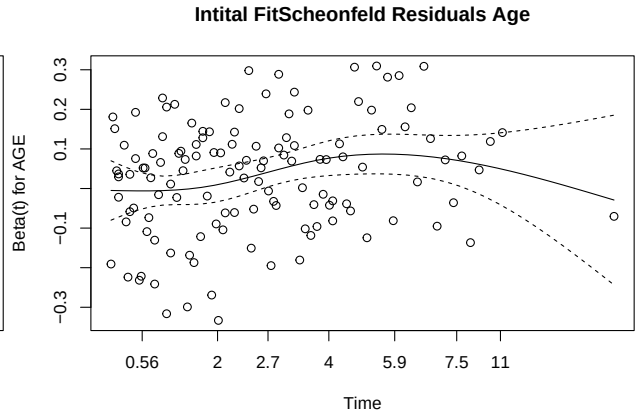


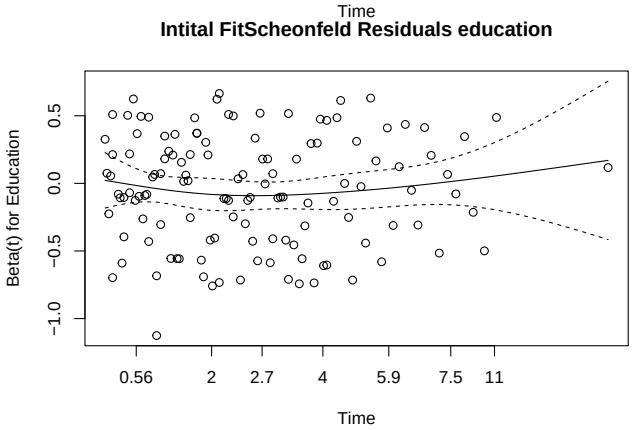


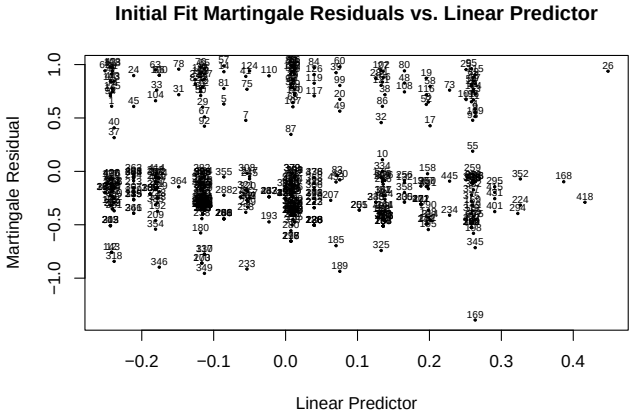


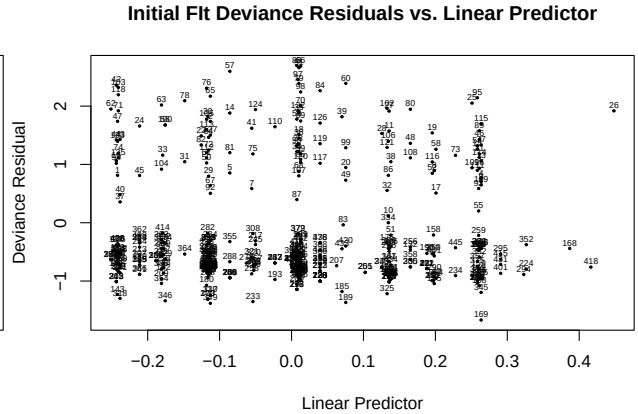
2.3 Initial Model Diagnostics

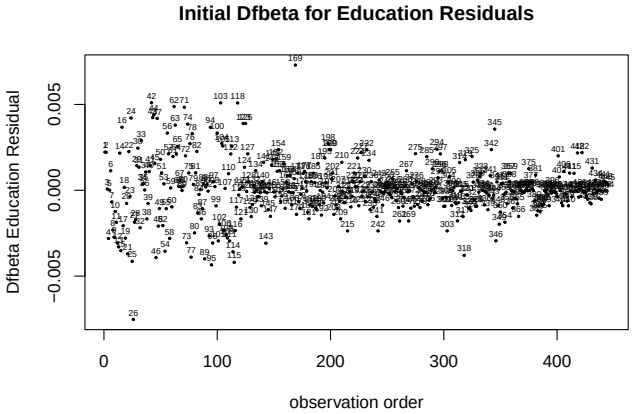




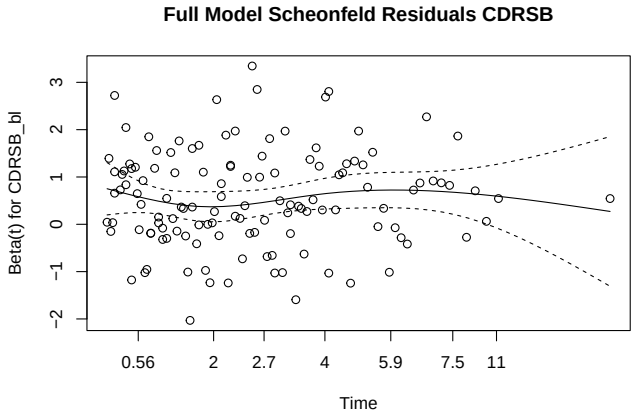


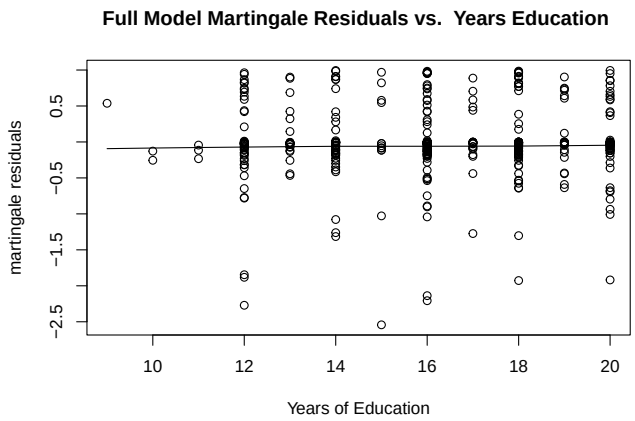


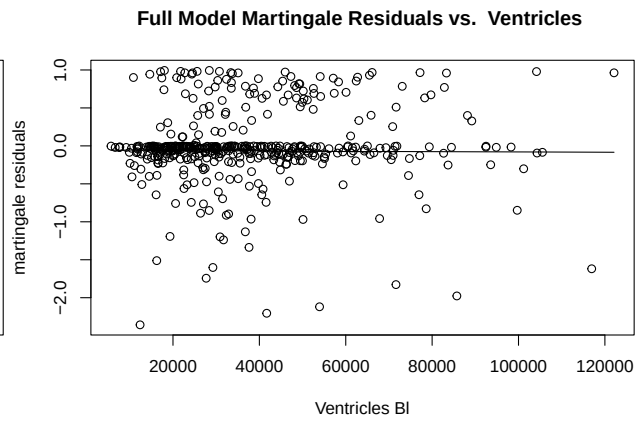


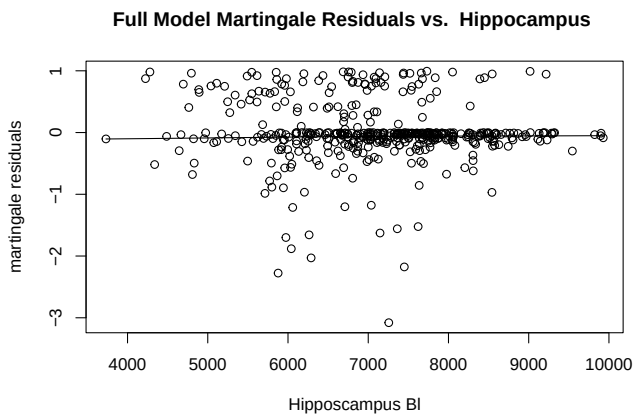


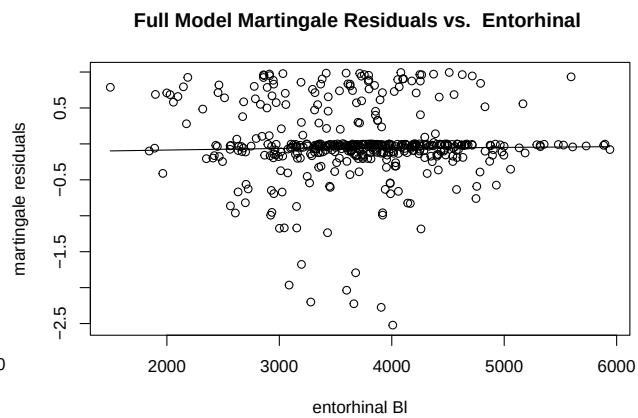
2.4 Full Model Diagnostics

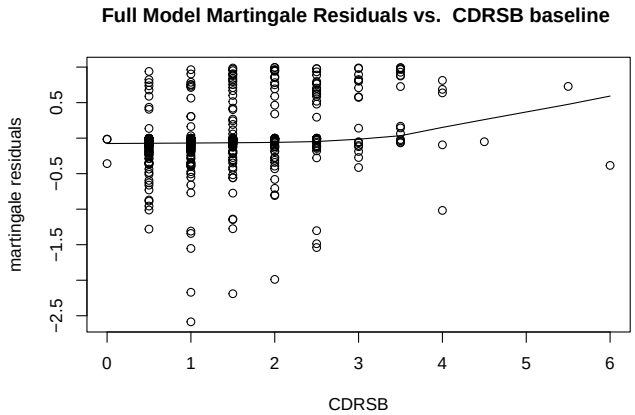


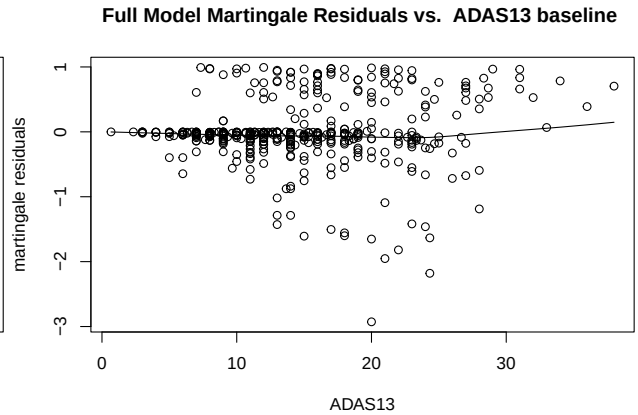


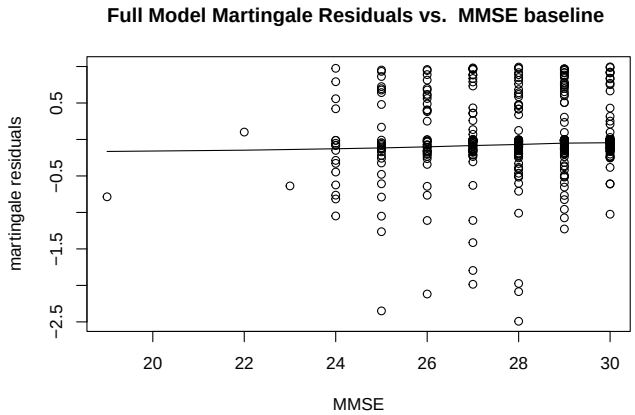


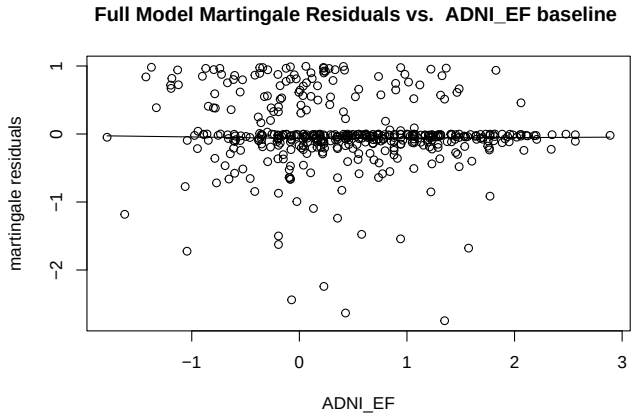


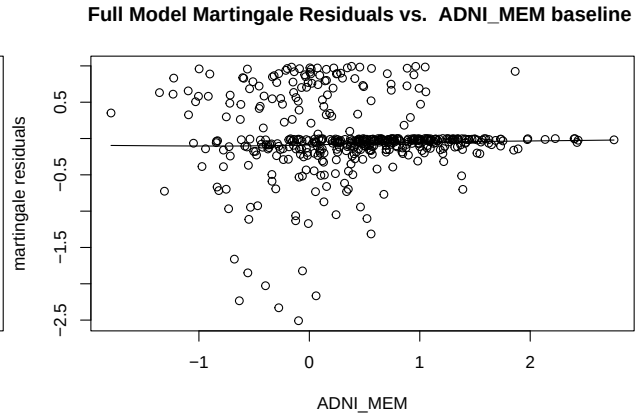


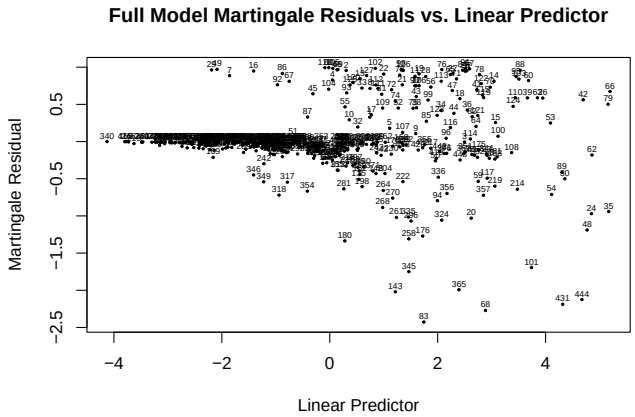

Full Model Martingale Residuals vs. CDRSB baseline
martingale residuals
CDRSB


Full Model Martingale Residuals vs. ADAS13 baseline
martingale residuals
ADAS13

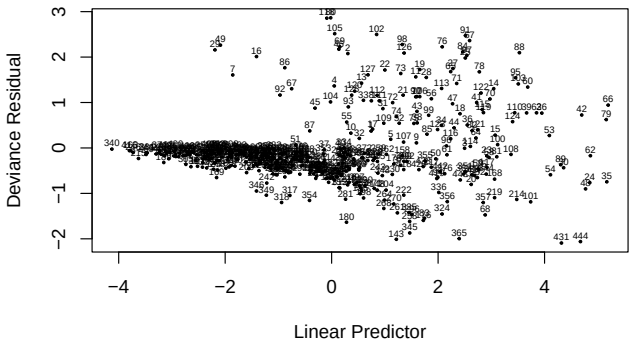

Full Model Martingale Residuals vs. MMSE baseline
martingale residuals
MMSE


Full Model Martingale Residuals vs. ADNI_EF baseline
martingale residuals
ADNI_EF

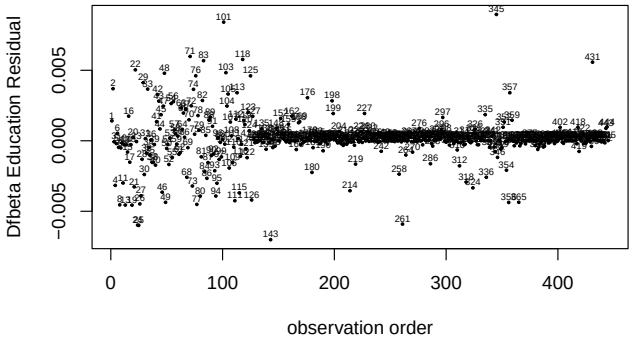

Full Model Martingale Residuals vs. ADNI_MEM baseline
martingale residuals
ADNI_MEM


Full Model Martingale Residuals vs. Linear Predictor
Martingale Residual
Linear Predictor

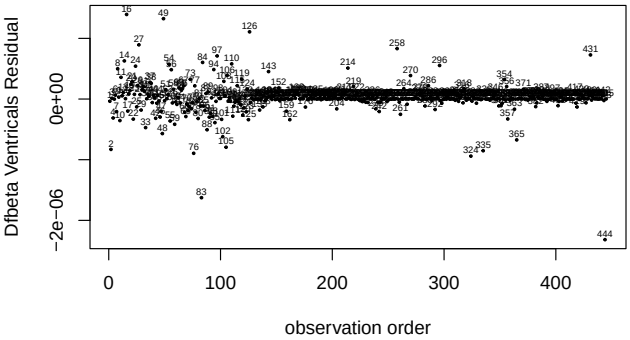
Full Model Deviance Residuals vs. Linear Predictor



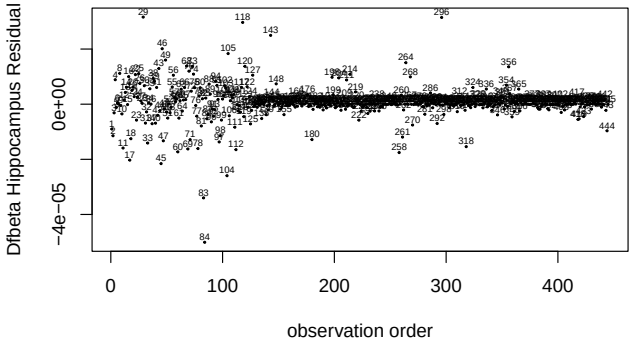
Full Model Dfbeta for Education Residuals



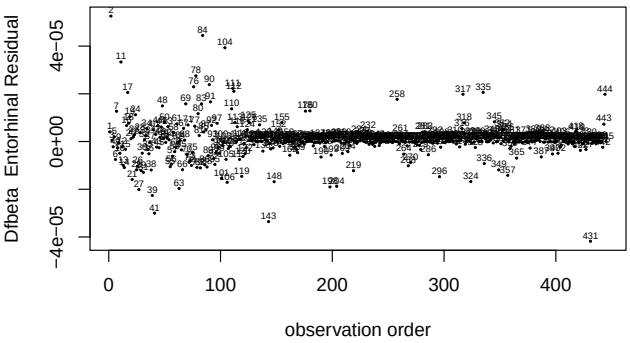
Full Model Dfbeta for Ventricals Residuals



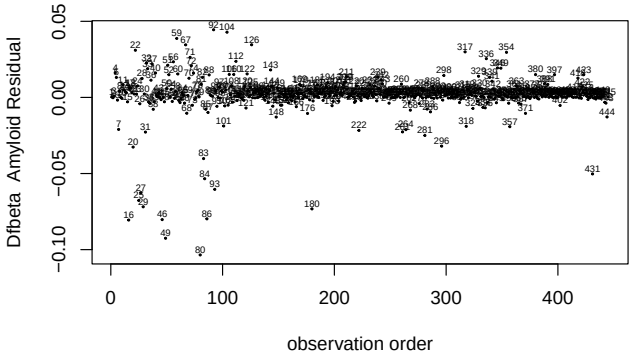
Full Model Dfbeta for hippocampus Residual



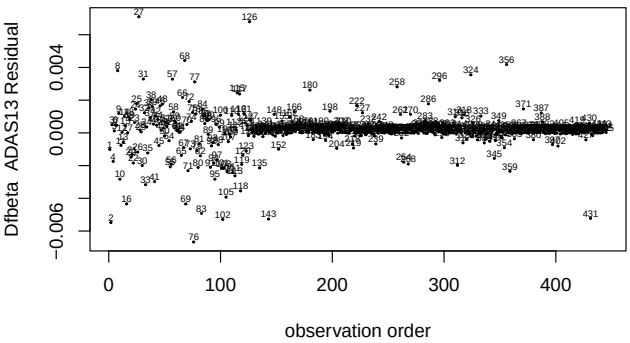
Full Model Dfbeta for Entorhinal Residual

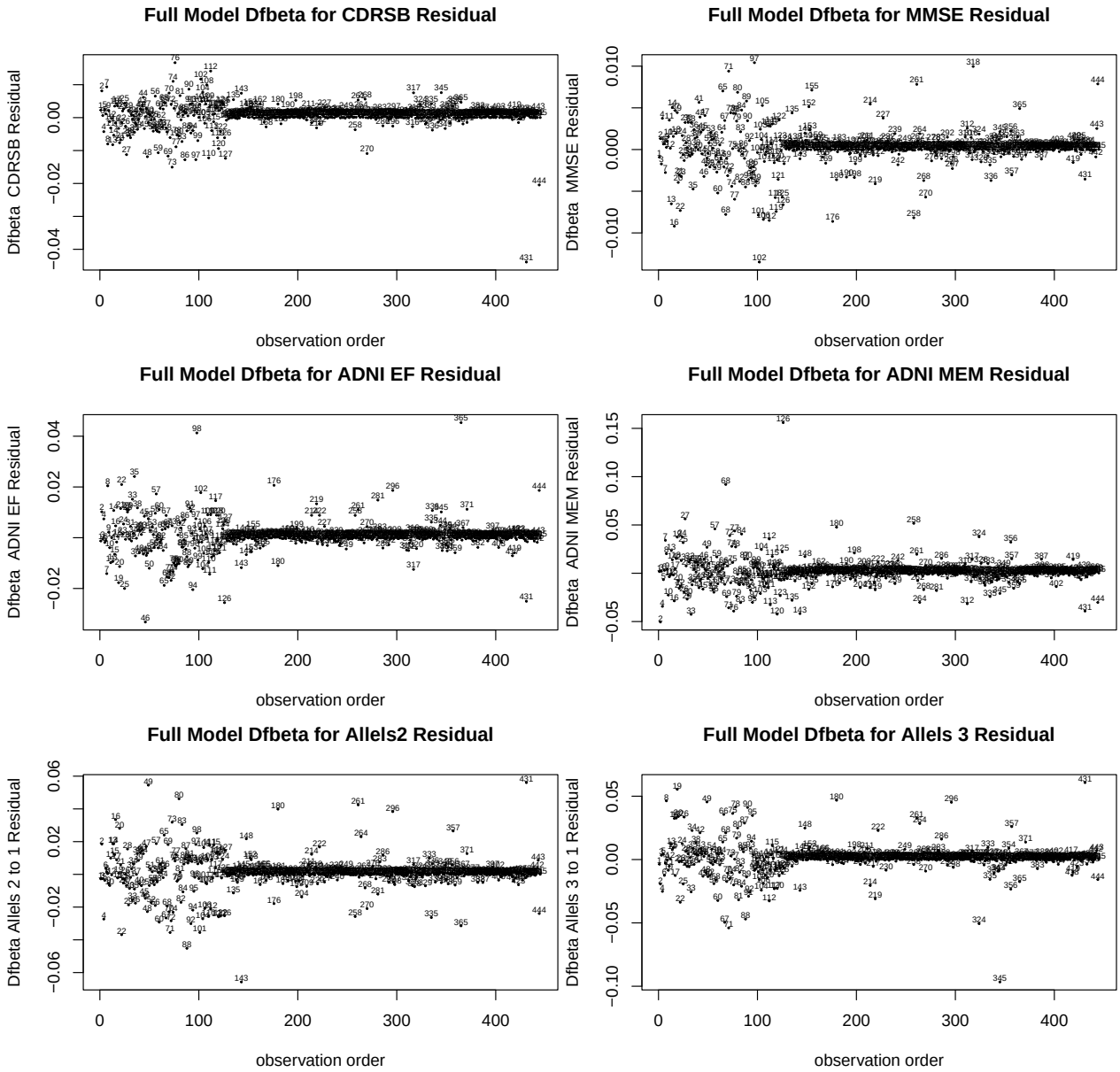


Full Model Dfbeta for Amyloid Residual



Full Model Dfbeta for ADAS13 Residual





2.5 Interesting Points

Table 10: Possible Outliers of Initial Model

	Education	Years_bl	D_status
26	9	1.00205	1
168	10	1.05407	0
169	12	12.13140	0
418	10	3.29637	0

examination of the martingale, deviance, and dfbeta residuals in the initial model reveals four

points of interest. participant 169 has an average education time and a long censored survival time. participant 26 has a relatively low education and a very short time to dementia. 168 and 418 have low education and short censored times.No obvious errors are present.

Table 11: Possible Outliers of Full Model

	Edu	CDRSB	ADAS	MMSE	Ventr.	Hippo.	Entor.	Years	EF	MEM	Demnt.
7	17	2.0	9.0	29	32379	7835.0	4254	7.8	0.9	0.2	1
16	20	1.5	10.0	25	122117	7436.0	4716	4.0	0.0	0.4	1
29	18	1.0	15.0	29	24517	9217.0	4627	7.3	0.6	1.2	1
49	15	1.5	8.0	29	104172	7661.0	4372	4.0	0.9	0.2	1
68	16	2.5	20.0	27	41699	6288.0	3280	4.2	-0.6	0.4	1
83	15	2.5	22.0	28	85725	7256.0	3663	6.4	-0.1	-0.1	1
84	16	3.5	18.0	29	77202	4279.0	3730	0.6	0.2	0.3	1
101	12	2.0	24.0	27	29241	7147.0	3679	2.0	-0.7	-0.2	1
126	16	2.0	10.7	29	66106	7519.4	3794	2.0	1.9	-0.3	1
143	20	1.0	23.0	28	53934	6263.0	3906	7.0	0.1	0.2	0
345	12	1.0	18.0	27	27684	5974.0	3197	6.0	-0.1	-0.2	0
365	18	1.5	21.0	26	71629	6041.0	3599	3.2	-0.4	-1.6	0
431	12	6.0	24.3	28	12380	7449.6	4009	3.2	-0.3	1.4	0
444	16	4.0	24.3	25	116944	5876.9	3087	1.0	-0.6	-1.0	0

The above 14 data points were selected for review from martingale, deviance, and dfbeta scatter plots. Paitient 7 has a low ADAS13 score, but longer uncensored survival time. Point 16 has a high education, lower ADAS13, very high Ventricle score with a medium uncensored survival time. participant 29 has a low CDRSB score and a longer uncensored survival time. participant 49 has a low CDRSB and ADAS13 score, large ventricle regions but fairly long uncensored survival time. participant 68 has no extreme scores and an average uncensored survival time. participant 83 has a large ventrical region, and a fairly long uncensored survival time. participant 84 has a high CDRSB score, low hippocampus value and a very short time till progression. participant 101 has a high ADAS13 score, very low ventrical value and high ADNI EF score but a short uncensored time till progression. participant 126 has a low ADAS13 score, high ADNI EF score, and short uncensored time till progression. participant 143 has a very high education, low CDRSB, high ADAS13 but a long censored survival time. participant 345 has a low CDRSB, ventricles, hippocampus, entorhinal, value aswell as negative ADNI scores but a fairly long censored survival time. participant 365 has very low ADNI EF score but medium censored survival time of .431 has a very high CDRSB score, high ADAS13 very low ventrical value and high ADNI EF but a medium censored survival time. participant 444 has a high CDRSB, ADAS, Ventrial score and a very short censored survival time.

Table 12: Risk Estimates without certain variables

Predictor	No MMSE B	No ADNI-Mem B	No MMSE e^B	No ADNI-Mem e^B	No MMSE e^B stdz	No ADNI-Mem e^B stdz	No MMSE SE	No ADNI-Mem SE
Edu	0.02	0.0132	1.0241	1.0133	1.0241	1.0133	0.0367	0.0365
Ventrc	0.00	0.0000	1.0000	1.0000	1.1387	1.1476	0.0000	0.0000
Hippoc	0.00	-0.0003	0.9997	0.9997	0.7527	0.7107	0.0001	0.0001
Entorh	0.00	-0.0002	0.9997	0.9998	0.8022	0.8394	0.0002	0.0002
Amtld+	1.74	1.7895	5.7210	5.9863	5.7210	5.9863	0.3201	0.3229
adas13	0.10	0.1158	1.0999	1.1227	1.8850	2.1605	0.0284	0.0191
cdrsb	0.54	0.5507	1.7160	1.7345	1.6676	1.6846	0.0941	0.0930
mmse	NA	-0.4688	NA	0.6257	NA	0.6257	NA	0.1434
EF	-0.42	0.0848	0.6592	1.0885	0.6592	1.1623	0.1455	0.0574
MEM	-0.19	NA	0.8303	NA	0.8303	NA	0.3033	NA
1 E4	0.19	0.2542	1.2052	1.2894	1.2052	1.2894	0.2332	0.2333
2 E4	0.73	0.7888	2.0738	2.2008	2.0738	2.2008	0.3144	0.3107

The above table shows the parameter estimates and errors for two models. Both models are attained by removing one predictor from the full model: either MMSE or ADNI-Mem. This comparison is made to investigate the unexpected sign of the MMSE coefficient in the final model. VIF did not reveal large amounts of collinearity in the covariates, but this is an additional exploration into the strange sign for MMSE. Here we see that Removing ADNI-Mem does not change the sign of the coefficient. Similar comparisons eliminating other single cognitive scores do not alter the sign of the MMSE coefficient.